



BY 534: Functional Neuroimaging and Data Analysis

Course Code: BY 534

Course Name: Functional Neuroimaging and Data Analysis

L-T-P-C: 2-1-0-3

Prerequisites: None

Students intended for: Open Elective course for B.Tech, M.Tech & PhD students

Elective or Core: Elective

Mutual Exclusion: -

Approval: 59th BoA

Course Contents

- **Basics of Programming:** Introduction to programming, Basic data types and structures, data input-output, functions, control statements, signal plotting, image visualisations, and basic statistics. [8 Lectures]
- **Neuroanatomy:** Anatomy of the brain, different brain lobes, cortical areas and their functions. [2 Lectures]
- **Designing Experiments to Study Brain Function:** Introduction to psychophysics, laws of psychophysics, stimulus design and considerations, data analysis, introduction to Psychophysics Toolbox. [2 Lectures]
- **Electro- and Magnetoencephalography:** The origin of EEG and MEG signals, basic preprocessing, artefact removal, independent component analysis (ICA) method for removing artefacts, evoked responses, spectral decomposition, time-frequency analysis, and multivariate pattern analysis (MVPA) methods for EEG and MEG. [8 Lectures]
- **Functional Magnetic Resonance (fMRI) :** Basics of Magnetic Resonance Imaging (MRI), different MRI techniques, an introduction to functional MRI (fMRI), blood oxygen level-dependent signal (BOLD), preprocessing methods, a general linear model (GLM), introduction to the statistical parametric mapping (SPM) toolbox, basics of brain connectivity. [8 Lectures]

Laboratory/practical/tutorial Modules: (14 Hours)

The students will attend one hour of lab per week. This will include hands-on experience in MATLAB programming, designing experiments to study brain functioning, and basic EEG and fMRI analysis.

Text books:

1. Cohen, M. X., **Analyzing Neural Time Series Data: Theory and Practice**, MIT Press, 2014.
2. Buxton, R. B., **Introduction to Functional Magnetic Resonance Imaging: Principles and Techniques**, 2nd Edition, Cambridge University Press, 2019.



References:

1. Sattar, T. A., **Introduction to MATLAB for Bioengineers**, Academic Press, 2019.
2. Lu, Z.-L., and Doshier, B. A., **Visual Psychophysics: From Laboratory to Theory**, MIT Press, 2013.